

MAT 3150 — Homework 3

Due September 24

Problem 1 (10 points)

Prove that for any $x, y \in \mathbb{R}$,

$$||x| - |y|| \leq |x - y|.$$

Proof. We start from the identity

$$x = (x - y) + y.$$

Taking absolute values and applying the triangle inequality,

$$|x| = |(x - y) + y| \leq |x - y| + |y|.$$

Subtracting $|y|$ from both sides gives

$$|x| - |y| \leq |x - y|.$$

We can do the same thing starting from the identity

$$y = (y - x) + x.$$

Taking absolute values and applying the triangle inequality,

$$|y| = |(y - x) + x| \leq |y - x| + |x|.$$

Subtracting $|x|$ from both sides gives

$$|y| - |x| \leq |y - x|.$$

Since $y - x = -(x - y)$ and $|-t| = |t|$ for every real t , we have $|y - x| = |x - y|$.

Now let $a = |x| - |y|$ and $c = |x - y|$. Note that $c \geq 0$. The first inequality says $a \leq c$. In the second, the left-hand side is $|y| - |x| = -a$ and the right-hand side is $|y - x| = c$, so it says $-a \leq c$, i.e. $a \geq -c$. Together,

$$-c \leq a \leq c.$$

Since $c \geq 0$, this is equivalent to $|a| \leq c$, that is,

$$||x| - |y|| \leq |x - y|,$$

which is the original statement.

□

Problem 2 (10 points)

Use the limit definition to compute

$$\lim_{n \rightarrow \infty} \frac{n+2}{3n+5}.$$

Claim. $\lim_{n \rightarrow \infty} \frac{n+2}{3n+5} = \frac{1}{3}.$

Proof. First we simplify the distance to $\frac{1}{3}$. Using the common denominator $3(3n+5) = 9n+15$,

$$\frac{n+2}{3n+5} - \frac{1}{3} = \frac{3(n+2) - (3n+5)}{3(3n+5)} = \frac{3n+6-3n-5}{9n+15} = \frac{1}{9n+15}.$$

Since this is positive for every $n \in \mathbb{N}$, and $9n+15 > 9n$,

$$\left| \frac{n+2}{3n+5} - \frac{1}{3} \right| = \frac{1}{9n+15} < \frac{1}{9n}.$$

Now let $\varepsilon > 0$. By the Archimedean property, choose $N \in \mathbb{N}$ with $N > \frac{1}{9\varepsilon}$, so that $\frac{1}{9N} < \varepsilon$. Then for all $n \geq N$,

$$\left| \frac{n+2}{3n+5} - \frac{1}{3} \right| = \frac{1}{9n+15} < \frac{1}{9n} \leq \frac{1}{9N} < \varepsilon.$$

Hence $\lim_{n \rightarrow \infty} \frac{n+2}{3n+5} = \frac{1}{3}.$

□

Problem 3 (10 points)

Prove that if $\lim_{n \rightarrow \infty} x_n = \infty$ and $(y_n)_n$ is bounded, then

$$\lim_{n \rightarrow \infty} (x_n + y_n) = \infty.$$

Proof. Since (y_n) is bounded, there is $M > 0$ with $|y_n| \leq M$ for every n , i.e. $-M \leq y_n \leq M$. In particular,

$$y_n \geq -M \quad \text{for every } n.$$

Recall that $x_n \rightarrow \infty$ means: for every $H \in \mathbb{R}$ there is $N \in \mathbb{N}$ such that $x_n > H$ for all $n \geq N$. Adding x_n to both sides of $y_n \geq -M$ gives, for every n ,

$$x_n + y_n \geq x_n - M.$$

So to make $x_n + y_n > K$, it is enough to make $x_n - M > K$.

Let $K \in \mathbb{R}$ be arbitrary. We want $x_n - M > K$; solving for x_n , this means $x_n > K + M$. Since $x_n \rightarrow \infty$, applying the definition with the height $H = K + M$ gives $N \in \mathbb{N}$ such that

$$x_n > K + M \quad \text{for all } n \geq N.$$

Then for all $n \geq N$,

$$x_n + y_n \geq x_n - M > (K + M) - M = K.$$

Since K was arbitrary, $\lim_{n \rightarrow \infty} (x_n + y_n) = \infty$.

□

Problem 4 (10 points)

Let $(x_n)_n$ be an increasing sequence of positive real numbers. Define

$$y_n = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

Prove that (y_n) is increasing as well.

Proof. We must show $y_n \leq y_{n+1}$ for every n , i.e. $y_{n+1} - y_n \geq 0$.

Let $S_n = x_1 + x_2 + \cdots + x_n$, so that $y_n = \frac{S_n}{n}$ and $S_{n+1} = S_n + x_{n+1}$. Using the common denominator $n(n+1)$,

$$y_{n+1} - y_n = \frac{S_{n+1}}{n+1} - \frac{S_n}{n} = \frac{n S_{n+1} - (n+1) S_n}{n(n+1)}.$$

We check the signs of the numerator and the denominator. The denominator satisfies $n(n+1) > 0$ since $n \geq 1$.

For the numerator, substitute $S_{n+1} = S_n + x_{n+1}$ and expand:

$$n S_{n+1} - (n+1) S_n = (n S_n + n x_{n+1}) - (n S_n + S_n) = n x_{n+1} - S_n.$$

Since (x_n) is increasing,

$$x_k \leq x_{n+1} \quad \text{for each } k = 1, \dots, n.$$

Adding these n inequalities, the left-hand sides sum to S_n and the right-hand side is n copies of x_{n+1} , so

$$S_n \leq n x_{n+1}, \quad \text{i.e.} \quad n x_{n+1} - S_n \geq 0.$$

Thus the numerator is nonnegative and the denominator is positive, so

$$y_{n+1} - y_n = \frac{n x_{n+1} - S_n}{n(n+1)} \geq 0.$$

Hence $y_n \leq y_{n+1}$ for every n , so (y_n) is increasing.

□